Sahasra Ranjan

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Research interests

Foundation models, AI agents, and robust generalization: long-horizon agent evaluation, knowledge retention, and multilingual and multimodal representation learning. Prior work spans code-switched NLP and protein-molecule interaction prediction; current work addresses real-time speech agents and synthetic evaluation environments.

Education

Indian Institute of Technology Bombay — Mumbai, India

B.Tech in Computer Science and Engineering | GPA: **9.05**/10 | Honors GPA: **9.75**/10 — *Aug 2019 - May 2023*

Honors & awards

- **O-1A Visa**, extraordinary ability in the sciences, granted by U.S. Citizenship & Immigration Services — *2026*
- **ACL 2023 Outstanding Paper Award**, co-first author; awarded to less than 1% of accepted papers — *2023*
- **Research Excellence Award**, best Bachelor's Thesis among graduating CS students, IIT Bombay — *2023*
- **JEE Advanced AIR 127** / 1.2M candidates; KVPY Fellowship; NTSE Scholar — *2017-2019*

Publications

*331 total citations, h-index 5, i10-index 4 (Google Scholar, Sept 2026). *denotes equal contribution.*

1. Richeek Das*, **Sahasra Ranjan***, Shreya Pathak, Preethi Jyothi. *Improving Pretraining Techniques for Code-Switched NLP.* **ACL 2023** Outstanding Paper Award
   Proposed boundary-aware masking strategies (Switch-MLM, Freq-MLM) that improved multilingual model performance on code-switched text by 5-7 F1 points across Hindi-English, Spanish-English, Tamil-English, and Malayalam-English benchmarks.
2. Alexander Kroll, **Sahasra Ranjan**, Martin K. M. Engqvist, Martin J. Lercher. *A general model to predict small molecule substrates of enzymes based on machine and deep learning.* **Nature Communications, 2023** *(248 citations)*
   Developed ESP, the first general enzyme-substrate prediction model, combining protein language models with molecular fingerprints; deployed as a public web server (esp.cs.hhu.de) used by researchers in drug discovery and metabolic engineering.
3. Alexander Kroll, **Sahasra Ranjan**, Martin J. Lercher. *A multimodal Transformer network for protein-small molecule interactions enhances predictions of kinase inhibition and enzyme-substrate relationships* (ProSmith). **PLoS Computational Biology, 2024** *(41 citations)*

Professional experience

Socratic AI Inc — Aug 2025 - Present

Founding AI Engineer (2nd engineer) — *Sunnyvale, CA*

- Designed and developed a synthetic-company simulation environment for **long-horizon evaluation of AI agents**, in which **21 simulated employees** generate email, chat, calendar, and spreadsheet activity on Google Workspace and Slack; agents are graded via checkpoint probes and LLM-based scoring, with activity calibrated to production telemetry.
- Built a **persistent memory system** for AI assistants, comprising an append-only conversation transcript store, source-grounded fact extraction, and two-stage nightly memory consolidation with rollback safeguards.
- Developed evaluation infrastructure for voice and workflow agents, including an LLM-as-judge and prompt-injection test harness, a tool-calling and latency benchmark across real-time models, and regression replay over **3,300+ captured production traces**.
- Architected a unified **real-time speech-to-speech inference layer** over Ultravox, Gemini Live, OpenAI Realtime, and PersonaPlex, covering turn-taking, voice activity detection, and barge-in; extended Gemini Live sessions beyond the 15-minute audio limit through context compression and session rollover.
- Implemented LLM cost controls that route high-cost requests to lower-cost models under per-user daily spend limits, with an audited organization-wide limit configurable at runtime.
- Led engineering, reporting to the founder, for the deployment at a publicly traded enterprise communications provider, one of the company's two **multi-year, multi-million-dollar contracts** (the other with a Fortune 150 semiconductor company); introduced provenance validation for AI-generated requirements documents after an internal audit.
- Delivered real-time SDKs in Python, TypeScript, and Swift, a Claude Agent SDK integration, and an Odoo ERP connector that reduced enterprise migrations from six weeks to days; authored **1,300+ commits** in the first year.

Mathkraft — 2024 - 2025

Co-founder, AI Math Tutoring Platform for K-8 — *San Francisco, CA*

- Co-founded an AI-powered math tutoring platform that embeds math problems in student-created stories and characters; piloted with **150+ students** at **95%+ retention** over six months.
- Managed a team of 4 engineers and 6 tutors and conducted 50+ in-person tutoring sessions to inform product design.

Rubrik Inc Summer 2022 (Intern), 2023 - 2024 (Full-time)
Software Engineer, Data Life Cycle Team *Bengaluru, India*

- Designed garbage collection for snapshot metadata, including reference tracking, a mark-and-sweep lifecycle, and gRPC APIs for detecting and removing orphaned entries across distributed storage nodes.
- Extended retention lock and legal hold to cloud-native workloads, enforcing immutability for petabyte-scale backups; implemented asynchronous SLA assignment using an event-driven poller across thousands of protected objects.

Research experience

ProSmith: Drug-Target Interaction via Multimodal Transformers Summer 2023
Research Collaborator | Prof. Martin J. Lercher, Alexander Kroll *HHU Düsseldorf, Germany*

- Extended enzyme-substrate prediction to drug-target interaction; designed the ProSmith architecture, combining bidirectional cross-attention between protein and molecule representations with gated fusion layers.
- Introduced contrastive pre-training on 500k BindingDB protein-ligand pairs, improving generalization to unseen drug scaffolds and out-of-distribution targets.
- Achieved **94.2% accuracy** on a held-out test set (30% error reduction over the prior state of the art) on kinase inhibition and enzyme-substrate benchmarks; deployed at deepmolecules.org, serving **50k+ predictions**.

Gradient-based Optimization for Continual Learning Jan 2023 - May 2023
B.Tech Thesis | Advisors: Prof. Preethi Jyothi, Prof. Manoj Prabhakaran *IIT Bombay*

- Developed **CommitteeSGD**, a continual-learning method that maintains a buffer of prior examples and modifies current gradients to penalize updates that conflict with learned knowledge; leveraged the local elasticity of neural networks (ICLR 2020) to limit decision-boundary perturbation.
- Achieved a **3.5% accuracy improvement** over standard SGD on CIFAR-100 (VGG16); periodic committee refresh outperformed a static buffer (**66.8%** vs. 63.4% test accuracy).

Improving Pretraining for Code-Switched NLP May 2022 - Dec 2022
B.Tech Thesis | Advisor: Prof. Preethi Jyothi, CSALT Lab *IIT Bombay*

- Showed that standard masked language modeling ignores language boundaries, leading multilingual models (mBERT, XLM-R) to learn spurious monolingual correlations on code-switched text.
- Designed **Switch-MLM**, a boundary-aware masking strategy that preferentially masks tokens at language switch-points to encourage cross-lingual dependencies.
- Introduced **Freq-MLM** for low-resource settings without language-ID labels, achieving a **5-7 F1 improvement** on NER and sentiment analysis across Hindi-English and Spanish-English benchmarks.

Deep Learning for Enzyme-Substrate Prediction Summer 2021
Research Intern | Prof. Martin J. Lercher, Alexander Kroll *HHU Düsseldorf, Germany*

- Built one of the first multimodal transformers for enzyme-substrate prediction, jointly encoding protein sequences (ESM-1b embeddings) and molecular structures (ECFP fingerprints) with learned attention.
- Replaced frozen ESM-1b embeddings with end-to-end fine-tuning, improving accuracy from **82% to 91%**.
- Curated 70k enzyme-substrate pairs from BRENDA, addressing severe class imbalance through negative sampling of biochemically plausible non-substrates.

Teaching

Teaching Assistant, CS725: Foundations of Machine Learning (Prof. Preethi Jyothi), IIT Bombay *Fall 2022*
Teaching Assistant, CS753: Automatic Speech Recognition (Prof. Preethi Jyothi), IIT Bombay *Spring 2023*

Technical skills

Languages: Python, C++, Go, TypeScript, SQL **Machine Learning:** PyTorch, Hugging Face Transformers, JAX
Infrastructure: Docker, Kubernetes, GCP, PostgreSQL **AI Systems:** Claude Agent SDK, RAG, vLLM, LLM-as-judge
Real-time: LiveKit, WebRTC, SIP, Gemini Live, OpenAI Realtime, Deepgram